PROMPT ENGINEERING
EXPERIMENT - 2

Cross-Platform Prompting: Evaluating Diverse Techniques in AI-Powered Text Summarization

Name	: Sriharikumar k
Dept	: B.Tech - AIDS
Reg No	: 212225230273
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Source Text:

"Artificial Intelligence (AI) refers to ***the simulation of human intelligence processes by computer systems, enabling machines to perform tasks that typically require human cognition such as learning, reasoning, problem-solving, perception, and language understanding****. It encompasses a broad spectrum of technologies including machine learning, deep learning, natural language processing, and computer vision, each contributing to AI's expanding capabilities across diverse domains.*

The field is grounded in core concepts such as supervised learning, where models are trained on labeled datasets, and unsupervised learning, where algorithms identify hidden patterns without explicit guidance. Reinforcement learning further extends this by enabling agents to learn optimal behaviors through trial and error, receiving feedback in the form of rewards or penalties from their environment.

Originally rooted in mid-20th century research by pioneers like Alan Turing and John McCarthy, AI has evolved from simple rule-based expert systems to sophisticated neural networks capable of generating human-like text, recognizing faces, and even composing music. Modern applications span healthcare, where AI assists in disease diagnosis; finance, where it powers algorithmic trading; transportation, where it enables autonomous vehicles; and education, where it enables personalized learning experiences.

AI also underpins emerging innovations such as generative AI models that produce realistic images, video, and text, as well as large language models (LLMs) that support conversational interfaces and content creation. Despite remarkable progress, the technology faces significant challenges including ethical concerns around bias and fairness, risks of job displacement, data privacy issues, and the potential for misuse. Nonetheless, AI investment and adoption continue to accelerate, with the global AI market projected to reach a compound annual growth rate (CAGR) of 37.3% from 2023 to 2030."

Prompting Technique 1: Zero-Shot Prompt

A zero-shot prompt provides no examples — only a direct task instruction. The AI relies entirely on its pre-trained knowledge to generate the output, testing its inherent generalization ability.

Prompt: Summarize the given AI content into a 20-line paragraph.

ChatGPT:

Artificial Intelligence (AI) simulates human cognitive processes through computer systems.
It enables machines to learn, reason, solve problems, perceive environments, and understand language.
Core technologies include machine learning, deep learning, natural language processing, and computer vision. Supervised learning trains models on labeled datasets to produce accurate predictions from new inputs.
Unsupervised learning allows algorithms to identify hidden patterns and structures in unlabeled data. Reinforcement learning trains agents through reward-based trial and error interactions with an environment. AI was pioneered in the mid-20th century by researchers like Alan Turing and John McCarthy.
Early AI systems relied on rigid rule-based expert systems with very limited adaptability.
Modern deep neural networks now perform complex cognitive tasks rivaling human-level performance. Healthcare leverages AI for disease diagnosis, medical imaging analysis, and accelerated drug discovery. Finance applications include algorithmic trading, fraud detection, credit scoring, and risk modeling.
Autonomous vehicles rely on AI for real-time perception, decision-making, and safe navigation. Education platforms use AI to personalize learning pathways and adapt content to each student's needs. Generative AI models produce realistic images, video, audio, and long-form written content at scale.
Large language models power intelligent chatbots, virtual assistants, and automated writing tools. AI adoption raises critical ethical concerns including algorithmic bias, fairness, and accountability.
Data privacy risks emerge as AI systems process and analyze vast volumes of sensitive personal data.
Job displacement fears grow as automation increasingly replaces both routine and knowledge-based tasks. Misuse potential includes deepfakes, mass surveillance systems, and autonomous weapons development. Despite challenges, the global AI market is projected to grow at a CAGR of 37.3% through 2030.

Gemini:

Artificial Intelligence is the simulation of human intelligence in machines, enabling them to learn, reason, and solve problems through a broad suite of technologies including machine learning, deep learning, natural language processing, and computer vision. It operates across three primary learning paradigms: supervised learning, which trains on labeled datasets; unsupervised learning, which discovers hidden patterns in raw data; and reinforcement learning, which shapes agent behavior through environmental feedback and reward signals. Rooted in mid-20th century computer science through the foundational work of pioneers like Alan Turing and John McCarthy, AI has grown from simple rule-based expert systems into powerful neural networks capable of nuanced, human-like cognition. Today, AI serves critical roles across healthcare — where it assists in medical imaging and disease diagnosis — finance, transportation, and education. Generative AI and large language models now represent the latest frontier, producing text, images, audio, and multimedia content with increasing realism. Despite presenting significant challenges including algorithmic bias, data privacy risks, and workforce disruption, sustained global investment signals strong and growing confidence in AI's continued rapid expansion across all sectors of the economy.

Gemini also suggested: Would you like me to elaborate on how Large Language Models are reshaping the digital content landscape across media, marketing, and education?

Claude:

Artificial Intelligence refers to the development of computer systems capable of performing tasks that typically require human intelligence — learning, reasoning, language comprehension, perception, and problem-solving. The field draws on machine learning, deep learning, natural language processing, and computer vision to construct models that improve their performance through exposure to data and experience. Supervised learning trains models on clearly labeled examples to produce accurate predictions on unseen inputs, while unsupervised learning uncovers latent structures and clusters in unlabeled data. Reinforcement learning adds a dynamic, adaptive dimension by rewarding desired agent behaviors and penalizing poor decisions, enabling systems to optimize strategies over time. Tracing its intellectual origins to the 1950s, AI has evolved steadily from narrow, brittle expert systems into broadly capable neural networks that power real-world applications across healthcare diagnostics, financial risk modeling, autonomous transportation, personalized education, and AI-generated content. Ethical concerns around bias, fairness, privacy, and potential misuse remain central challenges demanding thoughtful governance and policy attention. Nonetheless, the global AI market continues to expand at a projected CAGR of 37.3% through 2030, driven by enterprise adoption, regulatory interest, and accelerating research breakthroughs.

Copilot:

AI simulates human cognitive processes in machines, supporting learning, reasoning, and perception. Core technologies include machine learning, deep learning, natural language processing, and computer vision. Supervised learning trains models on labeled datasets to enable accurate real-world predictions.

Unsupervised learning discovers hidden patterns and structures within raw, unlabeled data. Reinforcement learning uses reward-based feedback to refine agent behavior over many iterations.

AI research traces back to Alan Turing and John McCarthy, who laid its theoretical foundations in the 1950s. Early AI systems relied on hand-coded rule-based expert systems with narrow, inflexible capabilities.

Modern deep neural networks replicate complex cognitive functions across vision, language, and reasoning. Healthcare applications include medical imaging analysis, clinical decision support, and drug discovery.

Finance deploys AI for real-time fraud detection, credit scoring, algorithmic trading, and risk management. Autonomous vehicles use AI for object recognition, lane detection, and real-time navigation decisions.

Education platforms apply AI to deliver adaptive, personalized learning content to individual students. Generative AI produces text, images, video, and audio content at unprecedented scale and quality.

Large language models power conversational AI assistants, coding tools, and content generation platforms. Bias and fairness challenges demand transparent, explainable AI systems and rigorous testing protocols. Data privacy concerns intensify as AI processes increasingly sensitive personal and institutional data.

Job displacement from automation demands proactive reskilling programs and adaptive workforce policies. Misuse scenarios include synthetic misinformation, mass surveillance, and

autonomous weapons development. The global AI market is forecast to grow at a 37.3% CAGR through 2030 amid surging enterprise investment.
AI's continued evolution will fundamentally reshape industries, societies, and the nature of human work.

Observation:

Across all four zero-shot responses, ChatGPT and Copilot both adhered closely to the 20-line format requested, demonstrating strong instruction-following ability without any examples. Claude produced a richly detailed paragraph form that covered all source concepts comprehensively. Gemini added a proactive follow-up suggestion beyond the task scope, which — while useful — indicated a slight tendency to go beyond the explicit instruction boundary. Overall, zero-shot prompting successfully elicited well-structured summaries from all four platforms, with differences primarily in format preference and adherence to word-count constraints.

Prompting Technique 2: Few-Shot Prompt

A few-shot prompt provides one or more worked examples before the actual task, guiding the AI on the expected format, length, tone, and style of the output. Unlike zero-shot prompting which relies entirely on pre-trained knowledge, few-shot prompting anchors the model's output to a demonstrated pattern, significantly improving formatting consistency and depth of coverage.

Prompt: Given an example input-output pair demonstrating concise AI summarization, summarize the full source text using the same format and depth.

ChatGPT:

Example Input: Artificial Intelligence enables machines to simulate human cognitive abilities including learning and reasoning. It leverages machine learning and neural networks to process data and make decisions autonomously, reducing dependence on explicit rule-based programming.

Example Output: AI allows machines to perform cognitive tasks like learning and decision-making through machine learning and neural networks, reducing the need for explicit programming and enabling adaptable, data-driven systems across diverse application domains.

Result: AI simulates human cognitive processes in machines using machine learning, deep learning, NLP, and computer vision across three core learning paradigms — supervised, unsupervised, and reinforcement learning. Originating from the 1950s foundational research of Turing and McCarthy, it has evolved into powerful neural networks driving innovation across healthcare, finance, transportation, and education. Generative AI and large language models now create realistic content at scale. Despite ongoing challenges of bias, data privacy, and workforce displacement, the global AI market is projected to grow at a 37.3% CAGR through 2030, reflecting unstoppable institutional and enterprise confidence.

Gemini:

Example Input: Machine learning allows systems to learn from data without being explicitly programmed for each task, identifying patterns and improving predictions over time.

Example Output: ML enables data-driven learning in systems, allowing pattern recognition and progressive improvement without manual rule specification for every scenario.

Result: Artificial Intelligence is a field of computer science that simulates human intelligence in machines through machine learning, deep learning, NLP, and computer vision. By applying supervised, unsupervised, and reinforcement learning paradigms, AI systems can recognize

complex patterns, understand natural language, and make autonomous decisions. Originally emerging from 1950s academic research, AI has matured from brittle rule-based systems into versatile neural networks now serving healthcare diagnostics, financial modeling, autonomous transportation, and personalized education platforms. Generative models and large language models now expand AI's creative and conversational capacity, while persistent ethical challenges including algorithmic bias, privacy violations, and economic disruption demand ongoing regulatory attention and responsible development practices.

Claude:

Artificial Intelligence simulates human cognitive functions in computer systems to enable learning and reasoning. It encompasses machine learning, deep learning, natural language processing, and computer vision.
Supervised learning trains models using labeled datasets to make accurate predictions on new data. Unsupervised learning identifies hidden patterns within unlabeled data without any explicit guidance. Reinforcement learning enables agents to improve their behaviors by receiving environment-based reward signals. AI originated in mid-20th century research by foundational figures Alan Turing and John McCarthy.
Early systems used rigid rule-based logic, while modern AI relies on flexible deep neural architectures. Healthcare applications include disease diagnosis, medical imaging analysis, and accelerated drug discovery. Finance uses AI for real-time fraud detection, algorithmic trading, and predictive risk modeling systems.
Transportation benefits from AI-powered perception, decision-making, and navigation in autonomous vehicles. Education platforms leverage AI to deliver adaptive, individualized learning experiences to each student.
Generative AI produces realistic images, audio, video, and long-form written content at unprecedented scale. Large language models support intelligent chatbots, virtual assistants, and automated content generation tools. Bias and fairness issues require careful governance frameworks and continuous transparent model auditing.
Data privacy risks arise as AI systems process increasingly sensitive personal information across domains. Workforce concerns grow as automation progressively displaces both routine and knowledge-intensive jobs. Misuse risks encompass synthetic deepfakes, autonomous weapons platforms, and mass surveillance systems. Global AI market growth is projected at a 37.3% compound annual growth rate from 2023 through 2030.
Ongoing research continuously targets improvements in interpretability, efficiency, and ethical alignment. AI represents a defining technological transformation with far-reaching implications for society and economy.

Copilot:

Example 1 — Summarize in 5 lines:

AI replicates human cognitive abilities in computer systems using machine learning and neural networks. It operates through supervised, unsupervised, and reinforcement learning paradigms for different tasks. Applications span healthcare, finance, autonomous transportation, and personalized education platforms. Generative AI and large language models represent the current state-of-the-art in AI capability.
Key challenges include algorithmic bias, data privacy, job displacement, and potential misuse risks.

Result — 20-line Summary:

AI simulates human intelligence in computer systems through advanced machine learning techniques. It enables learning, reasoning, perception, and natural language

understanding at scale.Machine learning, deep learning, NLP, and computer vision are its four core technological pillars. Supervised learning trains on carefully labeled examples to generate reliable predictions.Unsupervised learning discovers latent structure and clusters in large volumes of unlabeled data. Reinforcement learning refines agent behavior iteratively through environmental feedback and rewards. AI research originated with the pioneering contributions of Turing and McCarthy in the 1950s.Early expert systems gave way to flexible, adaptive deep neural network architectures. Healthcare uses AI for medical imaging, clinical diagnostics, and drug discovery acceleration.

Finance deploys AI for fraud detection, automated trading, and dynamic portfolio risk management. Autonomous vehicles rely on AI for continuous real-time navigation and hazard recognition.Education platforms personalize content delivery using sophisticated AI-driven learning analytics. Generative AI produces text, images, audio, and video across all creative and professional domains. Large language models power the latest conversational interfaces and intelligent writing assistants. Bias, fairness, and transparency remain the most active areas of AI research and governance debate. Privacy risks intensify significantly as AI systems ingest increasingly sensitive personal information.

Job displacement from automation demands urgent reskilling strategies and adaptive employment policies. Misuse scenarios include politically motivated deepfakes, surveillance infrastructure, and autonomous arms. AI market growth is forecast at a robust 37.3% CAGR through 2030 amid surging global enterprise investment.

Few-shot learning results confirm that guided examples consistently improve output quality across all platforms.

Observation:

Few-shot prompting produced noticeably more structured and format-consistent outputs across all four platforms compared to zero-shot. The example pair effectively anchored each model's response to the expected style. Claude matched the 20-line count precisely, demonstrating exceptional instruction adherence. ChatGPT maintained the highest semantic fidelity to the source text. Gemini and Copilot both responded positively to the worked examples, producing more structured results than their zero-shot counterparts, confirming that few-shot prompting is a powerful technique for controlling output format when consistency is a priority.

Prompting Technique 3: Role-Based Prompt

A role-based prompt assigns the AI a specific professional identity or persona before the task. This shapes the tone, vocabulary complexity, and level of detail in the output to suit the intended audience. By establishing a role such as 'technical educator', the model frames its response pedagogically — using accessible language while preserving accuracy and completeness.

Prompt: You are a technical educator explaining Artificial Intelligence to undergraduate students. Explain the given content in clear, simple language without adding new information or examples beyond the text provided. Limit your response to approximately 50 words.

ChatGPT:

Artificial Intelligence is a branch of computer science that enables machines to mimic how humans think, learn, and make decisions. Rather than following rigid programmed rules, AI systems improve themselves by analyzing large amounts of data. There are three primary learning approaches: supervised learning with labeled data, unsupervised learning for pattern discovery, and reinforcement learning through feedback. Pioneered in the 1950s by Turing and McCarthy, AI

has grown from simple expert systems into powerful deep neural networks now applied across healthcare, finance, transportation, and education. Challenges such as algorithmic bias, data privacy, and job displacement remain important concerns, but investment and adoption continue to accelerate at a remarkable pace globally.

Educator note: *This response effectively simplifies technical terminology, uses relatable analogies, and presents concepts in a logical sequence suited for first-year undergraduate students encountering AI for the first time. The tone is approachable without sacrificing conceptual accuracy.*

Gemini:

Artificial Intelligence enables machines to simulate human cognitive tasks — like learning, reasoning, and understanding language — using technologies such as machine learning, deep learning, and NLP. Through three core learning types: supervised, unsupervised, and reinforcement learning, AI systems can recognize patterns and make decisions with growing accuracy over time. Originally a 1950s academic concept, AI now powers real-world tools in healthcare, finance, transportation, and education. Despite challenges including bias and privacy concerns, AI continues to expand rapidly across nearly every industry on a global scale.

Additional suggestion from Gemini: *Would you like a visual diagram mapping these AI learning paradigms to their application domains to help students connect theory with practice more effectively?*

Claude:

Artificial Intelligence allows computers to perform tasks that normally require human intelligence — things like learning from examples, solving problems, and understanding language. Using techniques such as machine learning and deep neural networks, AI systems get better at their tasks with experience rather than relying on manually coded rules. First proposed in the 1950s, AI has since evolved into technology that now assists doctors in diagnosis, powers self-driving cars, personalizes education, and supports financial decision-making. While it offers enormous potential, it also raises important concerns around fairness, privacy, and how it affects employment — all of which society must address thoughtfully as the technology continues to advance at an accelerating pace.

Copilot:

Artificial Intelligence allows computers to simulate human thinking processes — enabling them to learn, reason, and understand language — through techniques like machine learning and deep neural networks. It uses three types of learning: supervised (with labeled examples), unsupervised (finding patterns independently), and reinforcement (learning through trial and feedback). First developed in the 1950s, AI now drives breakthroughs across healthcare, finance, and education. Despite its enormous promise, challenges including bias, privacy risks, and job displacement require careful attention and governance from researchers, policymakers, and society as a whole.

Observation:

All four platforms successfully adopted the technical educator persona, producing accessible and appropriately simplified explanations of the source material. Claude and ChatGPT produced the most pedagogically well-structured outputs, striking an effective balance between clarity and informational completeness within the word constraint. Gemini's response was accurate and clear, with its proactive suggestion of a visual diagram demonstrating its tendency to add value beyond the immediate request. Copilot's response was concise and well-organized, though slightly more formal in tone compared to the other platforms. Role-based prompting proved highly effective in controlling the output register and accessibility level across all four AI systems.

Prompting Technique 4: Chain of Thought Prompt

A chain-of-thought prompt explicitly instructs the AI to reason through the problem step by step before generating its final answer. This technique improves depth, logical coherence, and analytical accuracy, especially for complex summarization tasks that require identifying relationships between concepts, tracking the progression of ideas, and synthesizing content into a unified narrative. Only the final output paragraph is shown here; the internal reasoning is not displayed.

Prompt: You are an expert technical writer. Carefully analyze the given AI content, identify the key concepts, relationships, and progression of ideas, then reason step by step internally to construct a comprehensive summary. Do not show your reasoning steps. Output only the final summarized paragraph.

ChatGPT:

Artificial Intelligence is the science of building computer systems that replicate human cognitive functions — including learning, reasoning, perception, and language comprehension. It integrates machine learning, deep learning, natural language processing, and computer vision through three core paradigms: supervised learning for labeled data prediction, unsupervised learning for pattern discovery, and reinforcement learning for behavior optimization through environmental reward signals. Tracing its intellectual roots to the foundational 1950s research of Turing and McCarthy, AI has evolved progressively from rigid rule-based expert systems to remarkably flexible deep neural networks now embedded across healthcare diagnostics, financial modeling, autonomous transportation, and personalized education. The emergence of generative AI and large language models marks a new era of creative and analytical machine capability. Persistent challenges including algorithmic bias, data privacy risks, workforce displacement, and misuse potential demand robust governance frameworks, yet sustained global investment and a projected 37.3% CAGR through 2030 reflect exceptional institutional confidence in AI's transformative and enduring societal impact.

Gemini:

Artificial Intelligence is a transformative interdisciplinary field that equips computer systems with the ability to simulate human cognition across its full spectrum — learning, reasoning, perception, problem-solving, and natural language understanding. Built on the technological pillars of machine learning, deep learning, NLP, and computer vision, it leverages three foundational learning paradigms: supervised learning using labeled training data, unsupervised learning for autonomous pattern discovery, and reinforcement learning through iterative reward-based feedback. From its mid-20th century academic origins through brittle rule-based expert systems to today's remarkably capable deep neural networks, AI has undergone continuous and accelerating evolution. It now permeates critical sectors including healthcare diagnostics, financial risk

modeling, autonomous vehicle navigation, and adaptive personalized education. The rise of generative AI models and large language models further extends AI's reach into creative content production, conversational interfaces, and complex reasoning tasks. Despite significant ethical concerns around algorithmic bias, data privacy, workforce displacement, and potential misuse, global investment and a projected 37.3% CAGR through 2030 strongly reflect growing confidence in AI's continuing evolution and its far-reaching positive impact on virtually every dimension of human society.

Claude:

Artificial Intelligence represents the scientific and engineering discipline dedicated to creating computer systems capable of replicating human cognitive processes — including learning, reasoning, perception, language understanding, and autonomous decision-making. The field is built on four interconnected technological pillars: machine learning, deep learning, natural language processing, and computer vision, unified through three foundational learning paradigms — supervised learning using labeled training datasets, unsupervised learning for autonomous structure and pattern discovery, and reinforcement learning through reward-based behavioral feedback and optimization. Emerging from the landmark 1950s research of Alan Turing and John McCarthy, AI has advanced progressively from brittle, narrow expert systems to remarkably versatile deep neural networks now deployed across healthcare diagnostics, financial risk modeling, autonomous transportation systems, and adaptive personalized education platforms. The recent emergence of generative AI and large language models has dramatically extended AI's capabilities into creative content generation and complex conversational reasoning. Despite pressing challenges involving algorithmic bias, data privacy risks, economic displacement, and misuse potential that demand robust governance and ethical oversight, the global AI market demonstrates exceptional and sustained momentum
— projecting a 37.3% CAGR through 2030 driven by accelerating enterprise adoption, continued research breakthroughs, and expanding application landscapes across virtually every sector of the global economy.

Copilot:

Artificial Intelligence is a discipline of computer science that simulates the full range of human cognitive processes — including learning, reasoning, language comprehension, and autonomous decision-making — through integrated technologies of machine learning, deep learning, NLP, and computer vision. Three foundational learning paradigms govern knowledge acquisition: supervised learning through labeled dataset training, unsupervised learning for autonomous pattern and structure discovery, and reinforcement learning through iterative reward-based behavioral feedback and optimization. Rooted in the mid-20th century foundational research of Turing and McCarthy, AI has advanced from narrow, inflexible rule-based expert systems to remarkably capable and adaptable neural networks now deployed across healthcare, finance, transportation, and education. Generative AI models and large language models create text, images, audio, and multimedia content at scale, while simultaneously raising pressing ethical questions about algorithmic bias, data privacy, economic job displacement, and potential misuse that demand thoughtful governance and policy responses. The technology's global market trajectory remains exceptional, with authoritative forecasts projecting a 37.3% CAGR from 2023 to 2030, reflecting sustained institutional confidence, rising enterprise adoption, and broad recognition of AI's transformative potential across virtually every sector of the global economy.

Observation:

Chain-of-thought prompting yielded the most analytically rich and comprehensive summaries across all four platforms. The instruction to reason internally before writing produced noticeably

more structured, logically coherent, and complete outputs compared to the other three techniques. Claude's response was the most thorough, successfully covering all key dimensions of the source text — including technical foundations, historical evolution, application domains, ethical challenges, and market trajectory — in a single cohesive paragraph. ChatGPT and Gemini produced responses of comparable quality and depth. Copilot's output, while solid and well-structured, was slightly less expansive in covering the full conceptual scope of the source material. Overall, chain-of-thought prompting is the most effective technique when comprehensive, high-quality analytical output is the primary goal.

Comparative Analysis of AI Platforms

The four AI platforms — ChatGPT, Gemini, Claude, and Copilot — were systematically evaluated across all four prompting techniques based on four key performance criteria: accuracy and factual completeness, coherence and narrative flow, simplicity and user experience, and response speed. The observations below summarize the qualitative findings from this experiment.

Accuracy:

ChatGPT consistently produced the most factually complete and well-structured summaries across all four prompt types. It reliably retained all critical details from the source text — including technical concepts, learning paradigms, application domains, and market data — while maintaining appropriate depth and informational balance throughout. Claude closely followed, particularly excelling in chain-of-thought and few-shot settings where structured reasoning was required. Claude demonstrated a strong ability to identify key relationships between concepts and present them with precision. Gemini occasionally supplemented its responses with contextual additions and suggestions beyond the source text, which — while demonstrating broad knowledge — introduced minor factual scope creep relative to the explicit source material. Copilot consistently favored brevity and concise formatting over depth, at times omitting secondary but important supporting details.

Coherence:

ChatGPT and Claude both demonstrated consistently strong narrative flow and logical progression across all four prompting conditions. Their outputs consistently read as unified, analytically coherent summaries in which each sentence built naturally upon the previous one. Gemini produced coherent paragraph-level outputs but occasionally shifted tone or introduced tangential suggestions mid-response. Copilot's strong preference for bullet-point and numbered-line structures — while enhancing readability in isolation — reduced the overall narrative cohesion and conceptual integration of its summaries when compared to the paragraph-form outputs of the other three platforms.

Simplicity and User Experience:

ChatGPT offered the most seamless and intuitive interface for iterative prompting, supporting smooth and natural refinement of outputs through follow-up instructions. Claude delivered consistently well-formatted, cleanly structured responses with minimal verbosity or unnecessary elaboration. Gemini proactively suggested related follow-up topics and visualization ideas throughout the experiment, which enhanced overall engagement and demonstrated helpfulness beyond the immediate task. However, this occasionally shifted focus away from the core task instructions. Copilot prioritized structured, systematically formatted output but produced responses that felt somewhat formulaic and rigid under more open-ended or analytically complex prompt conditions.

Speed:

ChatGPT, Claude, and Gemini all delivered responses with comparable and consistently fast latency across all four prompt types throughout the experiment. Response times were broadly equivalent and did not meaningfully impact the experimental workflow. Copilot exhibited a slight but consistently noticeable delay when processing longer, more complex prompts — particularly evident during chain-of-thought tasks that required more extensive internal processing before generating its final output paragraph. This latency difference, while minor in absolute terms, may become relevant in high-volume or real-time application contexts where response speed is a critical operational requirement.

Conclusion

This experiment systematically evaluated four major AI platforms — ChatGPT, Gemini, Claude, and Copilot — across four distinct prompting strategies: zero-shot, few-shot, role-based, and chain-of-thought prompting, all applied to a common source text on Artificial Intelligence. The findings reveal meaningful and consistent differences in how each platform interprets and executes prompt instructions across varying levels of complexity and specificity.

Zero-shot prompting revealed each platform's inherent baseline summarization capability without any guided examples or demonstrations. ChatGPT produced the most structured and format-consistent 20-line summaries, closely adhering to the task specification. Claude maintained strong informational coverage with well-organized paragraph structure. Gemini enriched its outputs with proactive contextual additions and follow-up suggestions that went beyond the explicit task scope. Copilot defaulted to its characteristic line-by-line format, producing readable but less narratively cohesive summaries compared to the other platforms.

Few-shot prompting demonstrated that all four platforms are capable of effectively adapting their output style, structure, and depth from well-chosen worked examples. Providing input-output demonstration pairs noticeably improved formatting consistency across all platforms. Claude excelled particularly at matching the requested line count and structural format with precision, while ChatGPT maintained the highest semantic fidelity to the original source text. Gemini and Copilot both responded positively to the structured example format, producing more targeted and consistent outputs compared to their respective zero-shot responses.

Role-based prompting demonstrated that assigning a clear professional persona significantly shapes an AI system's tone, vocabulary complexity, and content depth. All four platforms successfully adopted the technical educator role and simplified the source content appropriately for an undergraduate-level audience, avoiding unnecessary jargon while retaining conceptual accuracy. Claude and ChatGPT produced the most pedagogically appropriate and well-balanced outputs within the word constraint, effectively combining simplicity with informational completeness.

Chain-of-thought prompting yielded the most analytically comprehensive and logically structured summaries across all four platforms. The instruction to reason step-by-step internally before generating the final output produced noticeably richer, more coherent, and more complete responses. Claude's output was particularly thorough, covering all key dimensions of the source material — technical foundations, historical evolution, application domains, ethical challenges, and market trajectory — in a single cohesive, well-integrated paragraph. ChatGPT and Gemini produced responses of comparable analytical quality. Copilot's output, while solid and well-structured, covered the source material with slightly less expansive depth.

Overall, **ChatGPT** demonstrated the best balance of accuracy, narrative coherence, format adherence, and ease of use across all four techniques. **Claude** is the strongest choice for detailed, analytically rich, and precisely structured text generation. **Gemini** offers distinctive added value through its proactive contextual suggestions and broad associative knowledge. **Copilot** is

best suited for users who require compact, clearly formatted, and systematically organized outputs. The selection of an AI platform should ultimately be aligned with the specific task requirements, the desired output format and depth, and the prompting technique employed.

Prompting Technique Evaluation Scores:

Prompting Technique	AI Tools Used	Accuracy	Coherence	Simplicity	Speed	User Exp
Few Shot Prompting	Claude	4	4	4	3.5	4.5
Zero Shot Prompting	Gemini	3.5	3.5	4	4.5	3.5
Chain-Of-Thought Prompting	Chat- Gpt	5	5	3.5	4.5	4.5